UC San Diego

Trilinear estimates for quadratic dispersive equations

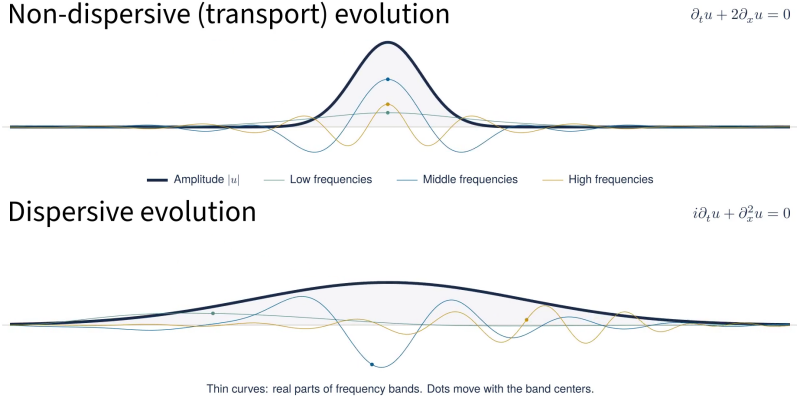
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includes work with Ioan Bejenaru and Tiklung Chan
University of California San Diego



UC Berkeley – HADES
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Introduction

What is dispersion?



Different frequencies move at different speeds, and the wave disperses!

Quadratic semilinear equations

Schematically, a quadratic and dispersive equation has the form

$$(i\partial_t + L(D))u = N(u),$$
$$u(0) = u_0$$

Examples: Schrödinger (Δ), Klein–Gordon ($\langle D \rangle$), wave ($|D|$).

Physics: optics, quantum mechanics, gravitation, water waves…

Motivating question

For which initial data do we get global-in-time solutions? How do these solutions behave as $t \to \infty$?

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Under what conditions do small initial data u_0 give rise to global-in-time solutions that asymptotically approach free waves?

$$\|u(t) - e^{itL(D)}u_*\|_{H^s_x} \longrightarrow 0 \quad \text{as} \quad t \to \infty.$$

where $u_{\text{free}}(t) = e^{itL(D)}u_*$ solves $(i\partial_t + L(D))u_{\text{free}} = 0$ with $u_{\text{free}}(0) = u_*$.

Why should one expect scattering?

$$(i\partial_t - \Delta)u = \underbrace{N(u)}_{\text{nonlinear forcing}}$$

Small accumulated forcing $\longrightarrow$ small correction to free evolution.

Linear dispersion

$$(i\partial_t - \Delta)u_{\text{free}} = 0$$

$$\|u_{\text{free}}(t)\|_{L^\infty_x} \lesssim (1+t)^{-n/2}.$$

Decay weakens quadratic forcing.

Possible ODE buildup

$$\partial_t v = v^2$$

$$v(t) = \frac{c}{T-t}.$$

Blows up as $t \longrightarrow T^-$.

> **Can dispersion win?** For small data, hope the late-time forcing becomes negligible and the solution approaches a free wave.

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Literature

Schrödinger equations

Germain–Masmoudi–Shatah (2009)
Bejenaru–Ionescu–Kenig–Tataru (2011)
Herr–Tataru–Tzvetkov (2011)
Ifrim–Tataru (2015)
Ifrim–Tataru (2023)
Ifrim–Tataru (2024)

Klein–Gordon equations

Shatah (1985)
Bejenaru–Herr (2017)
Chae–Oh (2017)
Ionescu–Pausader (2019)
Bejenaru–Borges (2026)
Borges–Chan (2026)

Wave equations and GR

Christodoulou–Klainerman (1993)
Tao (2001)
Tataru (2001)
Rodnianski–Tao (2004)
Lindblad–Rodnianski (2010)
Krieger–Sterbenz–Tataru (2015)
Krieger–Tataru (2017)
Ionescu–Pausader (2022)

Water waves

Wu (2011)
Germain–Masmoudi–Shatah (2012)
Alazard–Delort (2015)
Ionescu–Pusateri (2015)
Ifrim–Tataru (2016)
Deng–Ionescu–Pausader–Pusateri (2017)
Ionescu–Pusateri (2018)

Main result

Theorem (Bejenaru–B.–Chan)

The Schrödinger equation

$$(i\partial_t - \Delta)u = c_1 u^2 + c_2 \bar{u}^2, \qquad c_1, c_2 \in \mathbb{C}, \tag{2NLS}$$

with small initial data $u_0 \in \dot{H}^{-\frac{1}{2}}(\mathbb{R}^3)$ in the scaling-critical space is globally well-posed, scatters to free solutions, and has Lipschitz dependence on its initial data. The solution is also unique in the resolution space $\dot{X}^{-\frac{1}{2}} \subset C_t \dot{H}_x^{-\frac{1}{2}}$.

Roadmap & recap: multilinear estimates

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(v) Everything coming together

Contraction argument

$$(i\partial_t + L(D))u = F \qquad (F = N(u))$$

Goal: Find a complete space $X \subset C_t H^s_x$, and a forcing space Y such that

- If $\|u(0)\|_{H^s_x}$ is small, then $e^{itL(D)}u_0 \in X$
- Linear estimate: $\|u\|_X \lesssim \|u_0\|_{H^s_x} + \|F\|_Y$
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For small initial data, it is enough to establish

$$\|u\|_X \lesssim \|u_0\|_{H^s_x} + \|N(u)\|_Y, \quad \text{and} \quad \|N(u)\|_Y \lesssim \|u\|_X^p \tag{1}$$

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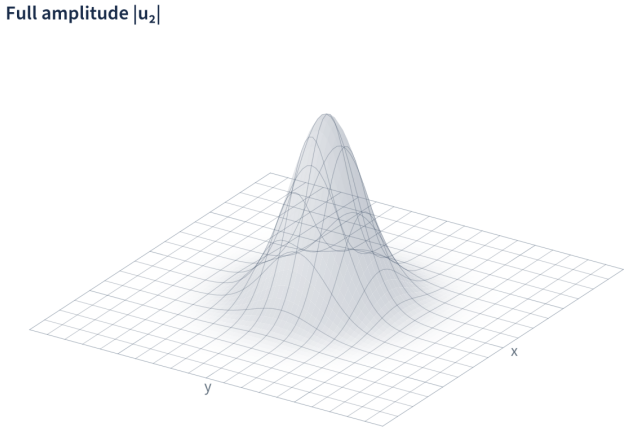
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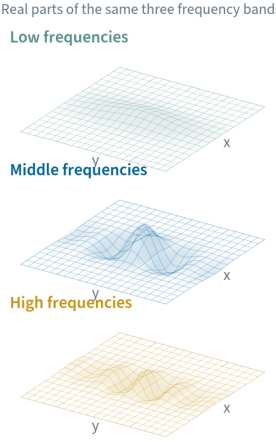
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In fact, if X is a *reasonable* space, then (1) establishes both global well-posedness and scattering.

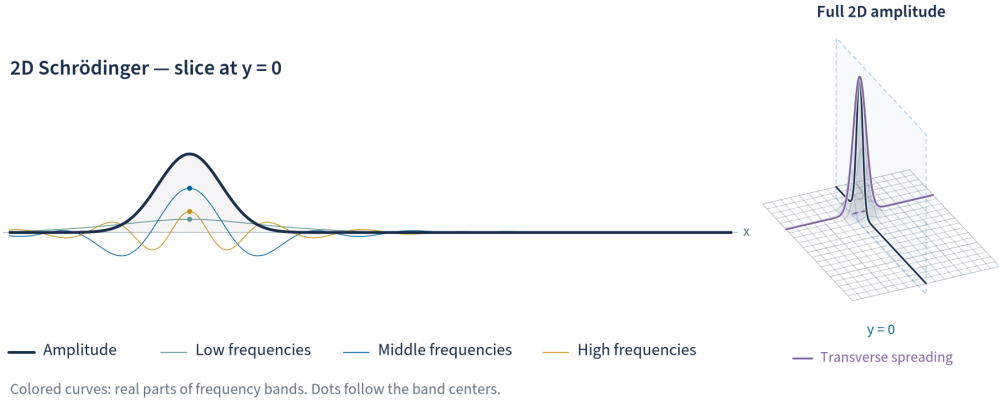
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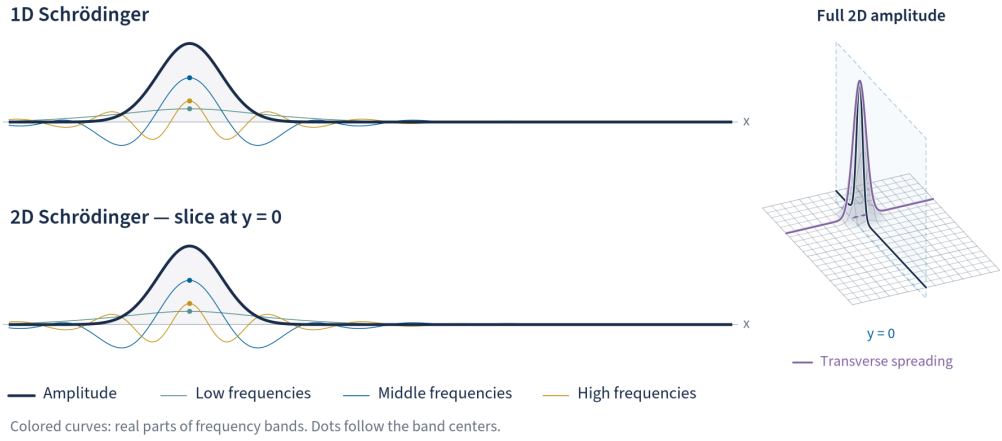
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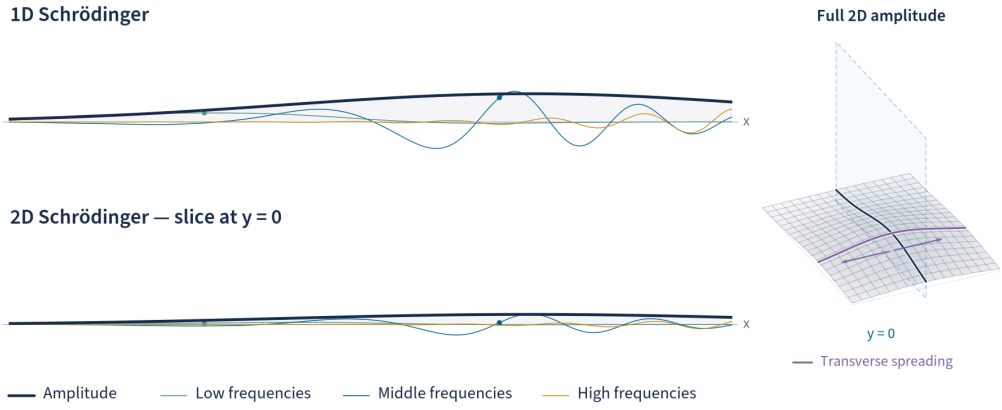
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Linear Schrödinger: Strichartz estimates

Free evolution: $\|e^{-it\Delta}u_0\|_{L_t^q L_x^r} \lesssim \|u_0\|_{L_x^2}, \qquad \frac{2}{q}+\frac{n}{r}=\frac{n}{2}, \quad 2 \le q, r \le \infty$

With forcing: if $(i\partial_t - \Delta)u = F$, then

$$\|u\|_{L_{t,x}^p} \lesssim \|u_0\|_{L_x^2} + \|F\|_{L_{t,x}^{p'}}, \qquad p = \frac{2(n+2)}{n}.$$

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Toy model: quadratic Schrödinger in dimension 4

$$(i\partial_t + \Delta)u = u^2, \quad u : \mathbb{R}_t \times \mathbb{R}^n_x \to \mathbb{C}.$$

Goal: Construct spaces X and Y such that $\|u^2\|_Y \lesssim \|u\|_X^2$.
By duality, it is enough to establish

$$|\langle u^2, w\rangle| \lesssim \|u\|_X^2 \|w\|_{Y^*}, \qquad w \in Y^*.$$

Strichartz estimates: $\|u\|_{L^p_{t,x}} \lesssim \|u_0\|_{L^2_x} + \|u^2\|_{L^{p'}_{t,x}}$, $p = \frac{2(2+n)}{n}$

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$$(i\partial_t + \Delta)u = u^2, \quad u : \mathbb{R}_t \times \mathbb{R}^n_x \to \mathbb{C}.$$

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First nonlinear iteration

> **Guiding principle:** nonlinear interactions carry more information than linear theory suggests

Consider the equation

$$(i\partial_t - \Delta)u = vw$$

where $v = e^{-it\Delta}g$, $w = e^{-it\Delta}h$ are free solutions localized at frequency 1.

In (spacetime) Fourier variables

$$\widehat{u}(\tau, \xi) = \frac{\widehat{(vw)}}{\tau - |\xi|^2}$$

This is well-defined when $\widehat{v} * \widehat{w}$ is supported *away* from the surface $\tau = |\xi|^2$. Such interactions are called *nonresonant*.

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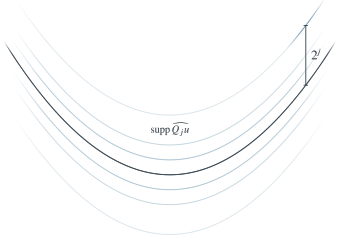
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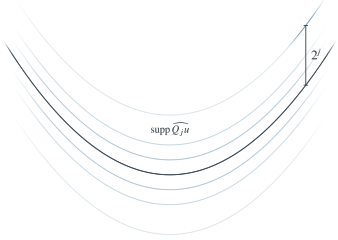
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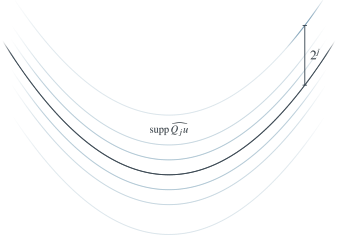
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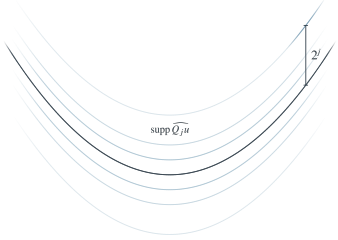
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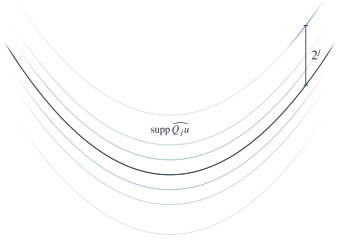
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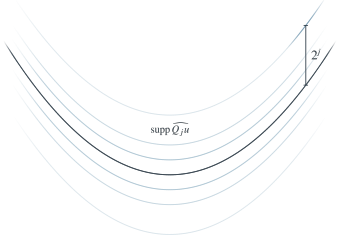
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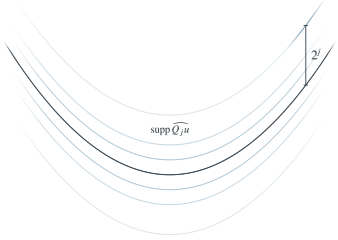
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Nonresonant Dirac–Klein–Gordon in 2D

Theorem (Bejenaru–B. ’26)

Let $M, m > 0$ with $2M > m > 0$. For small initial data

$$(\psi_0, \phi_0, \phi_1) \in H_x^{\frac{1}{2}+\epsilon}(\mathbb{R}^2) \times H_x^{1+\epsilon}(\mathbb{R}^2) \times H_x^{\epsilon}(\mathbb{R}^2),$$

the *nonresonant* system

$$(-i\gamma^\mu \partial_\mu + M)\psi = \phi\psi$$
$$(\Box + m^2)\phi = \langle \psi, \gamma^0 \psi \rangle$$

is globally well-posed and scatters.

Recap (trilinear formulation and resonance decomposition)

Our goal is to find spaces X and Y such that $|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}, \qquad w \in Y^*.$

- In high dimensions $n \geq 4$, Strichartz estimate on $L^3_{t,x}$ suffices.
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separately.
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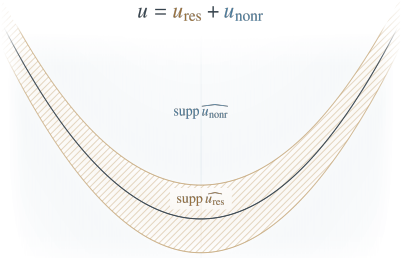
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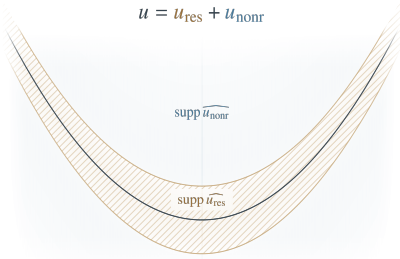


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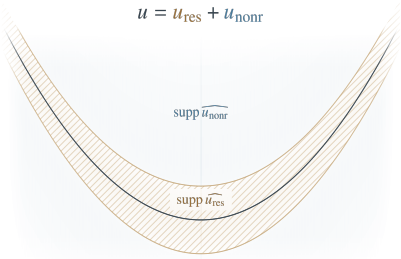


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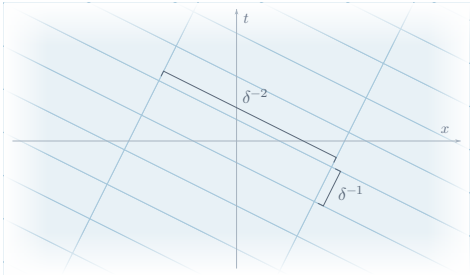
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$$\langle uv, w\rangle = \langle u_{\text{res}} v, w\rangle + \langle u_{\text{nonr}} v, w\rangle$$
separately.
- Since u_{nonr} is not a free solution, we have access to extra integrability.

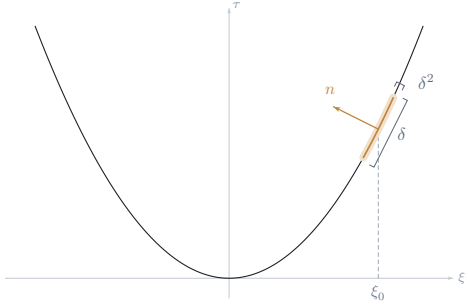


Since u_{res} is close to a free solution, we first consider the case where u, v, w are free waves.

Wave packets

Let $u = e^{-it\Delta} f$ be a free wave. In Fourier space, $\widehat{u}$ is supported on the paraboloid $\tau = |\xi|^2$:

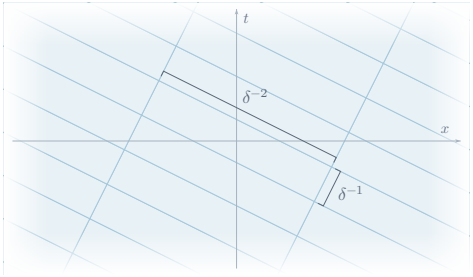


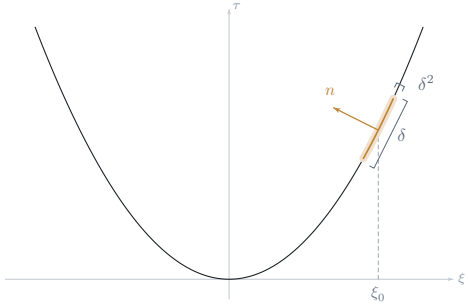


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Two-wave interactions

Question: In 3D, Hölder + Strichartz give

$$\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^{5/3}_{t,x}} \lesssim \|f\|_{L^2_x} \|g\|_{L^2_x}.$$

Can we do better?

If N_1 and N_2 are *transversal*, $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^p_{t,x}} \lesssim \|f\|_{L^2_x} \|g\|_{L^2_x}$, $\frac{3}{2} < p < \frac{5}{3}$

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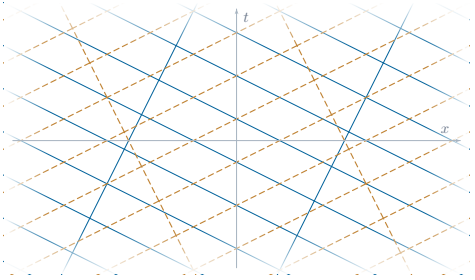
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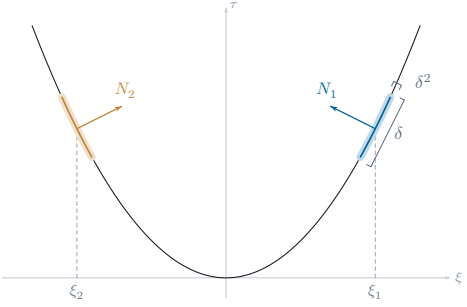
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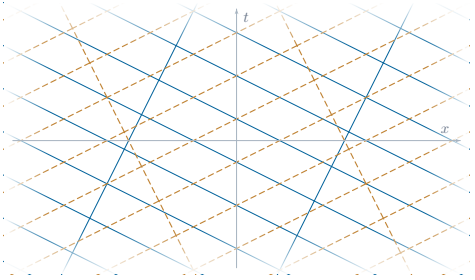
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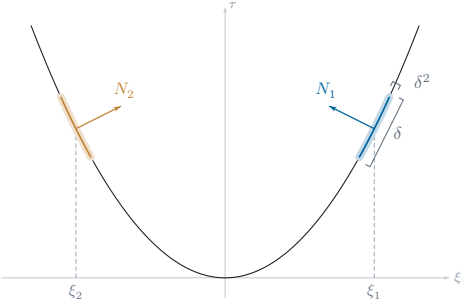
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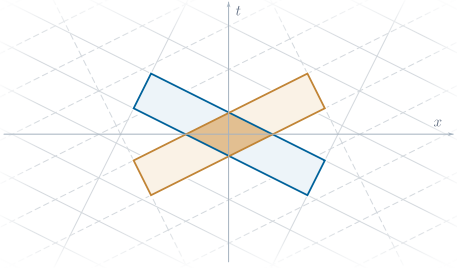


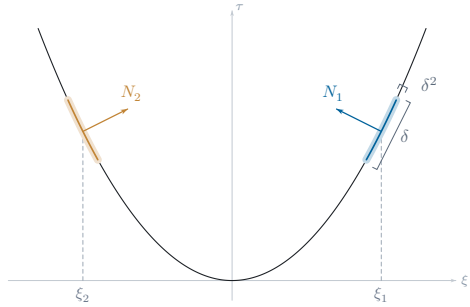


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Why is transversality important?

Heuristic: “transversal waves interact in smaller regions and for less time”



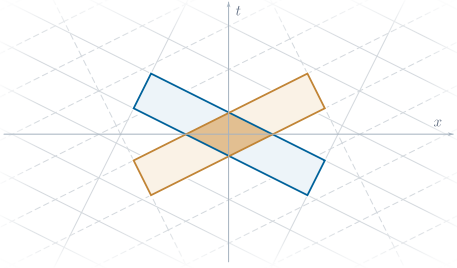


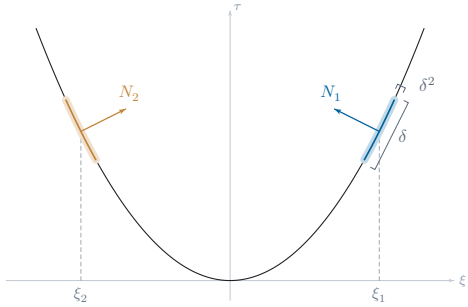
If the normals satisfy $|N_1 \wedge N_2| \geq \nu > 0$, then

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Bilinear estimates and the remaining resonances

For three free Schrödinger waves in dimension 3, the direct Hölder route is

$$\left|\int uv\bar{w}\ dx\,dt\right| \leq \|uv\|_{L^q_{t,x}} \|w\|_{L^{10/3}_{t,x}}, \qquad \frac{1}{q}+\frac{3}{10}=1.$$

We need a bilinear estimate with $q=\frac{10}{7}$, however bilinear restriction requires $q>\frac{3}{2}>\frac{10}{7}$

> **Takeaway:** the bilinear restriction range is not enough to control resonant quadratic interactions!

Bilinear suffices in the radial class: In that case, the correct Strichartz space is $L^{14/5}_{t,x}$, and so

$$\left|\int uv\bar{w}\ dx\,dt\right| \leq \|uv\|_{L^q_{t,x}} \|w\|_{L^{14/5}_{t,x}},$$

and the dual space $q=\frac{14}{9}>\frac{3}{2}$ belongs to the bilinear restriction range!

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The 3D Klein–Gordon–Schrödinger system

Theorem (B.–Chan, ‘26)

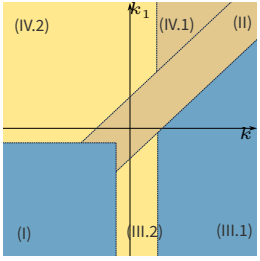
$$(i\partial_t + \Delta)u = un, \qquad (\Box + 1)n = \pm|u|^2.$$

For any $0 < \epsilon \ll 1$, sufficiently small **radial** data

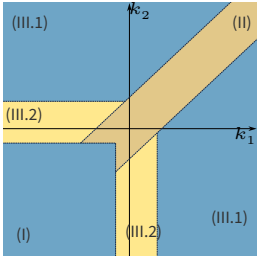
$$(u_0, n_0, n_1) \in L^2_x(\mathbb{R}^3) \times H^{-\frac{1}{2}+\epsilon}_x(\mathbb{R}^3) \times H^{-\frac{3}{2}+\epsilon}_x(\mathbb{R}^3)$$

give a global solution which scatters to free solutions as $t \to \pm\infty$.

Schrödinger equation



Klein–Gordon equation



Color key

- Blue: nonresonant
- Yellow: bilinear restriction
- Gold: radial

Can we do better than bilinear estimates?

Hölder tells you how to estimate $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^p_{t,x}}$ using only *linear* information.

Bilinear estimates use nonlinear information (transversality) to improve integrability $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^q_{t,x}}$ with some $q < p$.

Question: Do *trilinear* interactions capture any extra information compared to bilinear?

Yes!

$$\Big\| \prod_{j=1}^{3} e^{-it\Delta} f_j \Big\|_{L^p_{t,x}} \lesssim \prod_{j=1}^{3} \|f_j\|_{L^2_x}, \quad p > \frac{14}{15}$$

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Caveat: For general curves, transversality does not improve the range below $p = 1$.

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Transversality gives bilinear $L^2_{t,x}$.

Surfaces on $\mathbb{R}^{1+2}$

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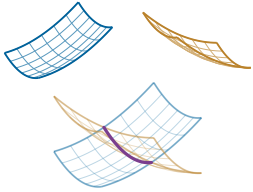
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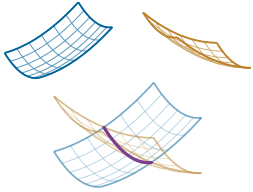
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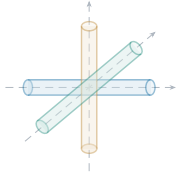
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(ii) Why do simpler approaches fail when studying (2NLS)?

(iii) Harmonic analysis: geometry of interacting waves

(iv) From harmonic analysis to PDEs

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Transference spaces

Recall that our original goal was to find spaces X and Y satisfying:

- Linear estimate: $\|u\|_X \lesssim \|u(0)\|_{L^2_x} + \|F\|_Y$,
- Multilinear estimate: $|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}$

We have discussed two types of multilinear estimates:

$$|\langle vw, u_{\text{nonr}}\rangle| \lesssim \|v\|_X \|w\|_X \|u_{\text{nonr}}\|_{L^2_{t,x}},$$

$$|\langle vw, u_{\text{res}}\rangle| \lesssim \|vw\|_{\text{Bil}} \|u_{\text{res}}\|_{\text{Bil}^*},$$

in the case where u_{res} is a free solution.

Question: How do we combine all this information into spaces X and Y?

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- Linear estimate: $\|u\|_X \lesssim \|u(0)\|_{L^2_x} + \|F\|_Y$,
- Multilinear estimate: $|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}$

We have discussed two types of multilinear estimates:

$$|\langle vw, u_{\text{nonr}}\rangle| \lesssim \|v\|_X \|w\|_X \|u_{\text{nonr}}\|_{L^2_{t,x}},$$

$$|\langle vw, u_{\text{res}}\rangle| \lesssim \|vw\|_{\text{Bil}} \|u_{\text{res}}\|_{\text{Bil}^*},$$

in the case where u_{res} is a free solution.

Question: How do we combine all this information into spaces X and Y?

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Candidate 1: Strichartz spaces – revisited

Let

$$X = L^2_t L^6_x \cap L^\infty_t L^2_x, \quad Y = L^2_t L^{6/5}_x + L^1_t L^2_x.$$

Since $Y^* = X$, the trilinear estimate reduces to

$$\Big| \int_{\mathbb{R}^{1+3}} uv\bar{w}\, dx\, dt \Big| \lesssim \|u\|_X \|v\|_X \|w\|_X$$

Pros

- Easy to define
- Easy to establish duality

Cons

- Only considers linear information
- Not enough to close the contraction when $n \leq 3$
- It does not allow for (bilinear) $L^2_{t,x}$

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$$e^{-it\Delta} f = \mathbb{1}_{[0,\infty)}(t) e^{-it\Delta} f$$



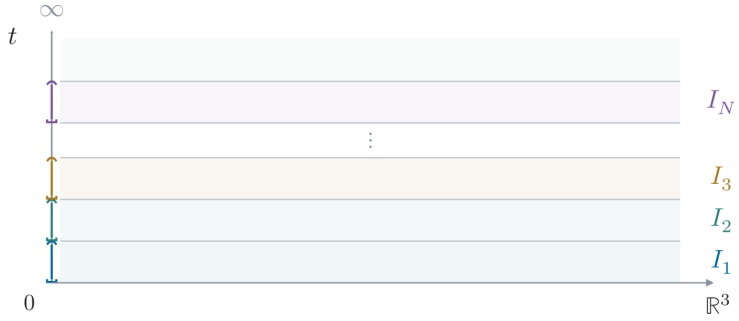
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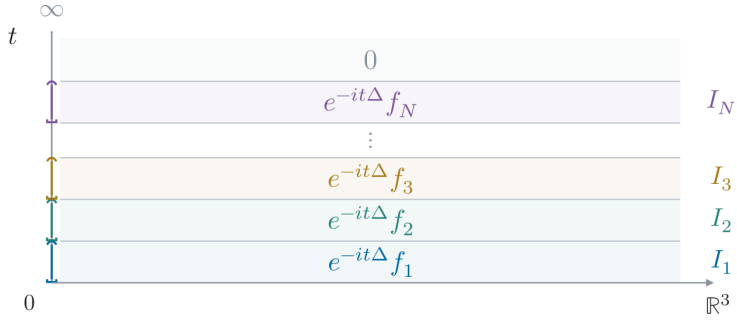
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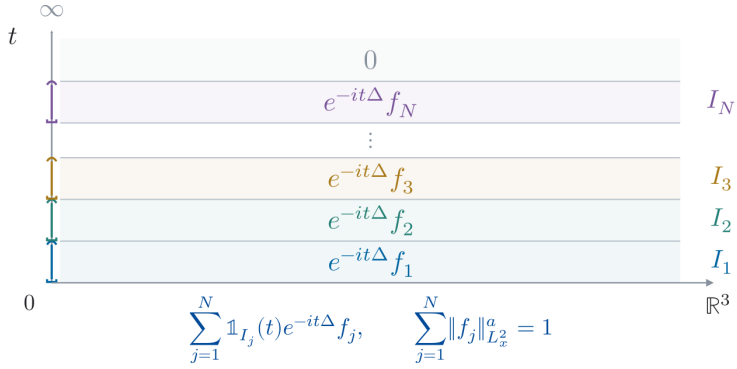
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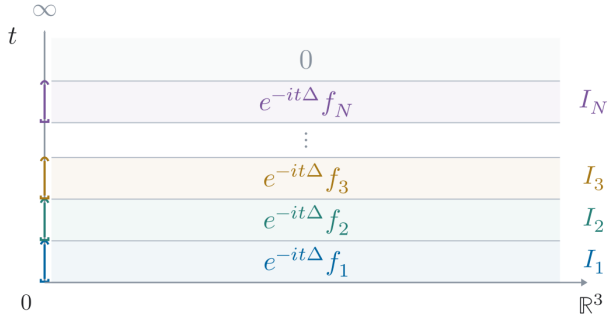
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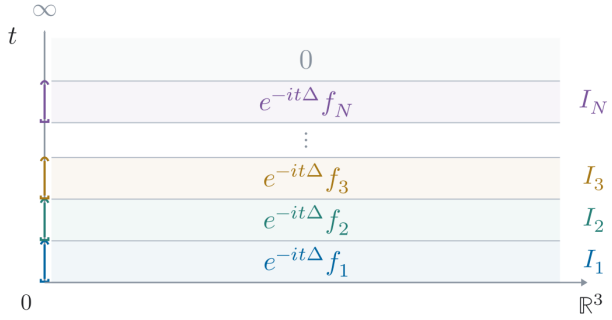
$$u = \sum_{j=1}^N \mathbb{1}_{I_j}(t) e^{-it\Delta} f_j$$

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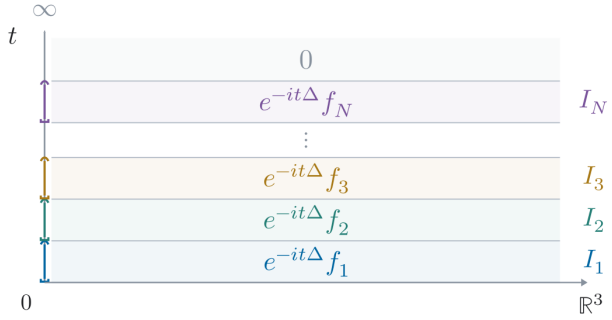
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$$U^a_\Delta = \left\{ u = \sum_{J=1}^\infty \lambda_J u_J : u_J \text{ is an atom, } (\lambda_J) \in \ell^1_J \right\}$$

Candidate 2: The atomic space U^2_Δ

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Transference from free solutions

U^2_Δ-functions are "almost free solutions" and most estimates for free solutions also hold in this space:

- Linear: if $\|e^{-it\Delta} f\|_{L^p_t L^q_x} \lesssim \|f\|_{L^2_x}$ for all $f \in L^2_x$, then $\|u\|_{L^p_t L^q_x} \lesssim \|u\|_{U^2_\Delta}$ for all $u \in U^2_\Delta$.
- Bilinear: if $\|e^{-it\Delta} f \cdot e^{-it\Delta} g\|_{L^q_{t,x}} \lesssim \|f\|_{L^2_x} \|g\|_{L^2_x}$, then $\|uv\|_{L^q_{t,x}} \lesssim \|u\|_{U^2_\Delta} \|v\|_{U^2_\Delta}$
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U^2_Δ is almost the perfect space... Except for duality.

$$|\langle uv, w\rangle| \lesssim \|u\|_X \|v\|_X \|w\|_{Y^*}, \qquad U^2_\Delta \subset V^2_\Delta$$

Estimate we need

$$\left| \int uv\bar{w}\, dx\, dt \right| \lesssim \|u\|_{U^2_\Delta} \|v\|_{U^2_\Delta} \|w\|_{V^2_\Delta}$$

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U^2_Δ-functions are "almost free solutions" and most estimates for free solutions also hold in this space:

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$$V_\Delta^2 = \left\{ e^{-it\Delta} v(t) : \sup_{\{t_j\} \text{ partitions}} \sum_j \|v(t_{j+1}) - v(t_j)\|_{L_x^2}^2 < \infty \right\}$$

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Goal: Explore the proof of scattering for $(i\partial_t - \Delta)u = u^2$

(i) The building blocks: contraction theorems and high-dimensional waves

(ii) Why do simpler approaches fail when studying (2NLS)?

- Slow dispersion in 3d, Insufficient time integrability
- Bilinear range is insufficient in general

(iii) Harmonic analysis: geometry of interacting waves

- Linear wave packets
- Bilinear range is very good in the radial case
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Main result: revisited

Theorem (Bejenaru–B.–Chan)

The Schrödinger equation

$$(i\partial_t - \Delta)u = c_1 u^2 + c_2 \bar{u}^2, \qquad c_1, c_2 \in \mathbb{C}, \tag{2NLS}$$

with small initial data $u_0 \in \dot{H}^{-\frac{1}{2}}(\mathbb{R}^3)$ in the scaling-critical space is globally well-posed, scatters to free solutions, and has Lipschitz dependence on its initial data. The solution is also unique in the resolution space $\dot{X}^{-\frac{1}{2}} \subset C_t \dot{H}_x^{-\frac{1}{2}}$.

In order to prove this, we need to:

- Classify the resonances
- Check whether resonant interactions are transversal
- Split $u = u_{\text{res}} + u_{\text{nonr}}$ and establish

$$\left| \int_{\mathbb{R}^{1+3}} u_{\text{nonr}} v \bar{w} \, dx \, dt \right| \lesssim \|u\|_{V_\Delta^2} \|v\|_{V_\Delta^2} \|w\|_{V_\Delta^2}, \qquad \left| \int_{\mathbb{R}^{1+3}} u_{\text{res}} v \bar{w} \, dx \, dt \right| \lesssim \|u\|_{V_\Delta^2} \|v\|_{V_\Delta^2} \|w\|_{V_\Delta^2}$$

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Resonances for $(i\partial_t - \Delta)u = u^2$

$$\int_{\mathbb{R}^{1+3}} uv\bar{w}\,dx\,dt = \left(\hat{u} * \hat{v} * \hat{\bar{w}}\right)(0) = \int \hat{u}(\xi,\tau)\hat{v}(\eta,\tilde{\tau})\overline{\hat{w}}(\xi+\eta,\tau+\tilde{\tau})\,d\xi\,d\eta\,d\tau\,d\tilde{\tau}$$

Resonant interactions occur exactly when u, v, and w are free waves, which are (Fourier) supported on the paraboloid $\tau = \xi^2$. So resonances occur when $\xi^2 + \eta^2 = (\xi+\eta)^2$, that is, when $\xi \cdot \eta = 0$.

To check transversality, we compute $|N_\xi \wedge N_\eta \wedge N_{\xi+\eta}| \approx \min\{|\xi|, |\eta|\}$, which degenerates at low frequencies!

Resonances for $(i\partial_t - \Delta)u = u^2$

$$\int_{\mathbb{R}^{1+3}} uv\bar{w}\,dx\,dt = \left(\hat{u} * \hat{v} * \hat{\bar{w}}\right)(0) = \int \hat{u}(\xi,\tau)\hat{v}(\eta,\tilde{\tau})\overline{\hat{w}}(\xi+\eta,\tau+\tilde{\tau})\,d\xi\,d\eta\,d\tau\,d\tilde{\tau}$$

Resonant interactions occur exactly when u, v, and w are free waves, which are (Fourier) supported on the paraboloid $\tau = \xi^2$. So resonances occur when $\xi^2 + \eta^2 = (\xi+\eta)^2$, that is, when $\xi \cdot \eta = 0$.

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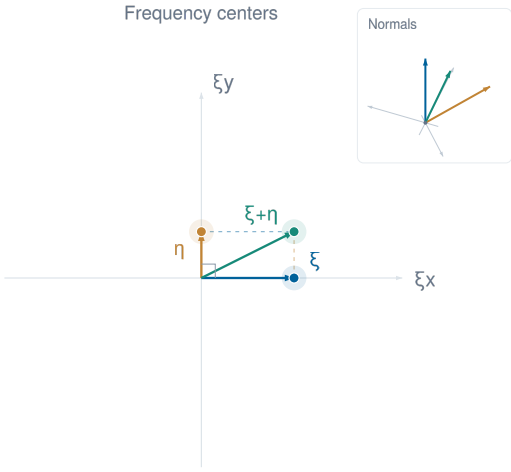
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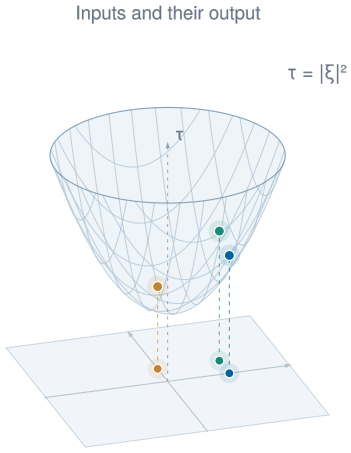
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Three-wave interactions





Nonresonant estimates

We can assume that v, w have frequency 1 and u has frequency 2^k, with $k \leq 0$.
Let $Q_j u$ localize u to the region $|\tau - \xi^2| \approx 2^j, j \in \mathbb{Z}$.

High modulation on u

High modulation on v

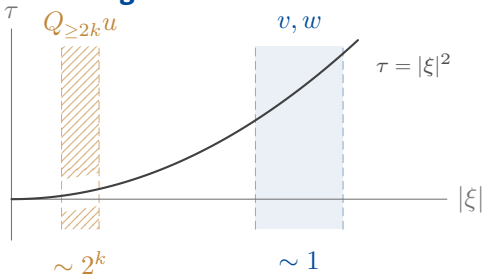
Schematic frequency–modulation slices; hatched regions have $|\tau - |\xi|^2| \gtrsim 2^{2k}$.

Claim: $\left|\int_{\mathbb{R}^{1+3}} (Q_{\geq 2k}u) v \bar{w}\, dx\, dt\right| + \left|\int_{\mathbb{R}^{1+3}} u (Q_{\geq 2k}v) \bar{w}\, dx\, dt\right| \lesssim \langle k \rangle^3 \|u\|_{V^2_\Delta} \|v\|_{V^2_\Delta} \|w\|_{V^2_\Delta}$

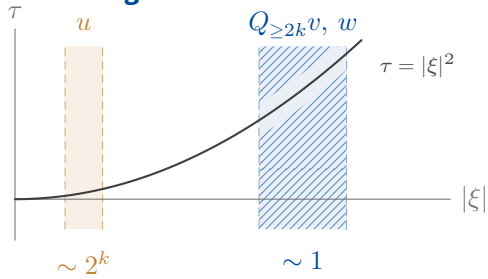
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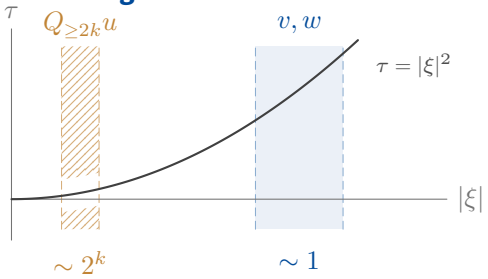
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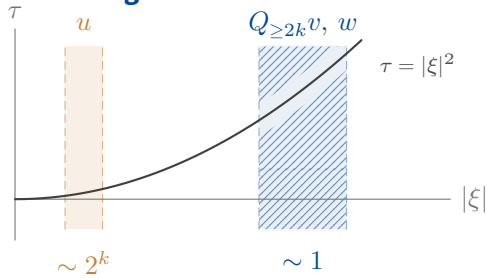
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Techniques: bilinear $L^2_{t,x}$, $\dot{X}^{0,1/2}$, Fourier cube decomposition

Resonant estimates

Goal: $\left| \int_{\mathbb{R}^{1+3}} Q_{<2^k} u \cdot Q_{<2^k} v \cdot \overline{Q_{<2^k} w}\, dx\, dt \right| \leq C_\epsilon 2^{\epsilon k} \|u\|_{V^2_\Delta} \|v\|_{V^2_\Delta} \|w\|_{V^2_\Delta}$

Step 1: Establish the free wave estimate

$$\left| \int_{\mathbb{R}^{1+3}} e^{-it\Delta} f \cdot e^{-it\Delta} g \cdot \overline{e^{-it\Delta} h}\, dx\, dt \right| \leq C_\epsilon 2^{-k\epsilon} \|f\|_{L^2_x} \|g\|_{L^2_x} \|h\|_{L^2_x}$$

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V^2_Δ does not have an atomic structure, and it is hard to work with directly. So instead we show that

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for some $\frac{14}{15} < p < 1$ and $a > 2$. Recall that $U^2_\Delta \subset V^2_\Delta \subset U^a_\Delta$, and so V^2_Δ-transference follows from interpolation.

The fact that we can prove an estimate for $p < 1$ is crucial for the interpolation, which means that the curvature of the paraboloid plays a key role in this result.

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Full recap

Goal: Explore the proof of scattering for $(i\partial_t - \Delta)u = u^2$

(i) The building blocks: contraction theorems and high-dimensional waves

(ii) Why do simpler approaches fail when studying (2NLS)?

- Slow dispersion in 3d, Insufficient time integrability
- Bilinear range is insufficient in general

(iii) Harmonic analysis: geometry of interacting waves

- Linear wave packets
- Bilinear range is very good in the radial case
- Transversality improves integrability
- Trilinear transversality is better than bilinear transversality

(iv) From harmonic analysis to PDEs

- Strichartz spaces are good for linear transference
- U^2_Δ is great for multilinear transference
- U^2_Δ has bad duality
- V^2_Δ has good duality, but transference is harder

(v) Everything coming together

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(v) Everything coming together

- Resonance analysis
- Check that trilinear interactions are transversal
- Use interpolation to transfer free estimates to V^2_Δ

TL;DR

- Multilinear wave interactions give contain extra information beyond the linear theory
- This information comes from transversality and curvature properties of the characteristic surface
- Using multilinear estimates beyond free solutions requires an transferring them to larger function spaces
- V^2_Δ is a very suitable transference space in a lot of cases

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Thank you!

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